GENIUS HACKS Q3 2026 · RISK ASSESSMENT WORKBENCH

FCRM Agent Blueprint

Option A, expanded into ten buildable modules: a deep multi-agent graph that carries a change request from intake to a committee decision, with two points nothing can cross without a human saying so.

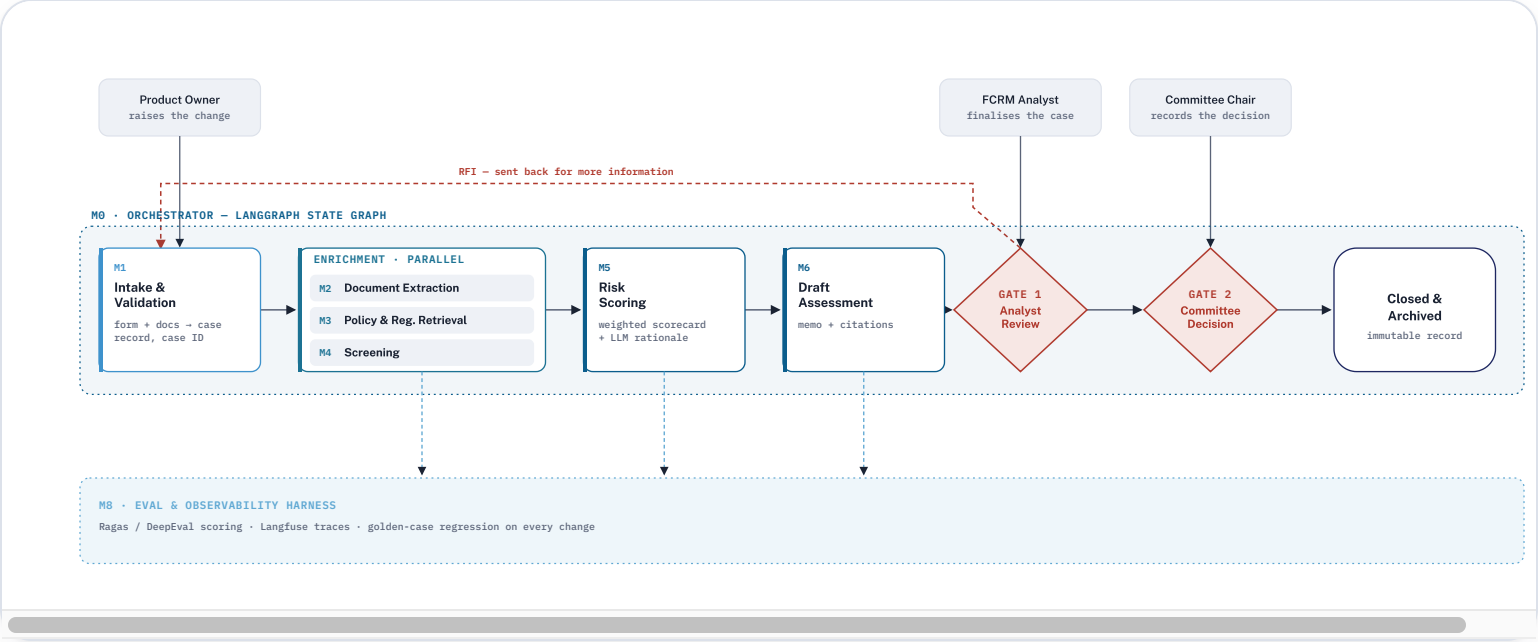
INTAKE Form + document uploads

POLICY Synthetic library, authored in-house

SCREENING Included, simulated data

COMMITTEE Single recorded decision

1 Draft → 2 Submitted → 3 Enrichment → 4 Draft assessment → 5 Analyst review · Gate 1 → 6 Committee review → 7 Decision recorded · Gate 2 → 8 Closed & archived



The orchestrator (M0) sequences intake, three parallel enrichment agents, scoring and drafting inside one LangGraph state graph. Gate 1 and Gate 2 are hard stops — nothing proceeds past either without a recorded human decision — and Gate 1 can send a case back to intake as a request for information. The evaluation harness samples every AI-bearing stage on each run.

Dev 1 · Agent Orchestration & AI Harness

Dev 2 · Domain, Data & Retrieval

Dev 3 · Application, Integration & Deployment

Dev 4 · Evaluation, Governance & Observability

Human review gate

Ten modules, each with one job

Every box in the diagram above expands to a module with a stated purpose, inputs and outputs, and a suggested tool. Build to these boundaries and the seams between developers

outputs, and a suggested tool. Build to these boundaries and the seams between developers stay clean.

M0 Dev 1

Orchestrator

Sequences every module call for a case, holds both gates open until a person acts, and manages the request-for-information loop back to intake.

In Case events: submitted, module-completed, decision-recorded

Out Next-step directive; case status written to the Case Database

`LangGraph` `interrupt() / resume` `Temporal (fallback)`

Carries the single heaviest judging weight — agent orchestration is 30% of score. The interrupt/resume pattern at each gate is the artifact that proves nothing self-approves.

M1 Dev 3

Intake & Validation

Validates the product owner's submission, assigns a case ID, classifies the change type, and flags incomplete fields before the case enters enrichment.

In Form: product type, change type, geography, segment, target date

Out Validated case record, or a needs-more-information event

`Pydantic / Zod` `Claude Haiku` `Postgres`

M2 Dev 2

Document Extraction

Parses uploaded product specs, vendor contracts, marketing copy and process diagrams into normalized case facts: parties, jurisdictions, customer types, transaction types.

In Uploaded PDFs, DOCX, images

Out Case-facts JSON, each field linked to its source page

`Claude (document)` `LlamaParse` `unstructured.io`

M3 Dev 2

Policy & Regulatory Retrieval

Retrieves the specific policy sections, risk-rating criteria and comparable past cases that apply to this change — not the whole library.

In Case facts, change classification

Out Ranked policy citations + precedent cases, each with a source

`pgvector` `Voyage embeddings` `Cohere rerank`

M4 Dev 2 + Dev 3

Screening

Runs simulated sanctions, PEP, adverse-media and vendor-risk lookups against synthetic reference data for every named party, and triages ambiguous matches.

In Party / vendor / geography list from case facts

Out Screening report: hit / no-hit / needs-review, with confidence

`Mock REST service` `Synthetic watchlists` `Claude Haiku (triage)`

M5 Dev 1

Risk Scoring

Produces an auditable risk tier from case facts, citations and screening results with a transparent weighted scorecard — the model explains the number, it never invents it.

In Case facts, policy citations, screening report

Out Risk tier + factor-by-factor breakdown (weight, value, contribution)

`Versioned scorecard (Python)` `Claude (rationale)`

M6 Dev 1

Draft Assessment

Synthesizes facts, citations, screening and score into the memo an FCRM analyst actually reviews: rating, citations, open questions, suggested conditions.

In Outputs of M2 through M5

Out Structured draft memo with inline citations

`Claude (long-context)` `Versioned memo template`

M7 Dev 1 + Dev 3

Human Review Gates

Two points the orchestrator cannot cross without a recorded action: Gate 1 the analyst finalises or sends an RFI; Gate 2 the chair records Approved, Rejected, Deferred, or Approved with Conditions. Every action lands in the audit log with actor and timestamp.

In Draft assessment (Gate 1); committee outcome (Gate 2)

Out Case advances, or returns to intake as an RFI

`LangGraph interrupt()` `Approvals table`

M8 Dev 4

Eval & Observability

Regression-tests extraction, retrieval, scorecard consistency and memo faithfulness against a golden-case set on every change, and traces every run for later reconstruction.

In Agent traces, golden-case set, live case outputs

Out Per-module eval scores; searchable trace log per case ID

`Ragas / DeepEval` `Langfuse` `CI regression job`

M9 Dev 3

Case & Notification

Tracks the SLA clock against the two-day target, notifies each persona at every transition, and is the system of record the audit log and Workbench UI both read.

In Case state transitions from the orchestrator

Out Status notifications; SLA timers and breach flags

`Postgres` `Templated email/webhook stub`

Data stores

Four stores back the modules above — all synthetic, none touching a real system.

Case Database

System of record for every case, its state and its approvals. Read and written by M1, M9, the orchestrator

Synthetic Policy Library

Authored FCRM policies, risk-rating criteria and procedures that M3 retrieves against.

Synthetic Screening Reference Data

Mock sanctions, PEP, adverse-media and vendor-risk records M4